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The Cosmological Constant as Inverse Holographic Information Capacity

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Abstract — We derive the cosmological-constant energy density ρ_Λ from the modular structure of the algebra of observables $\mathcal{A}(\mathcal{D})$ on the cosmic causal diamond, with no free parameters and no fine-tuning. The total horizon stress-energy density of the modular state evaluates to $\rho_{\text{info}}(z) = 3c^2 H^2(z)/(8\pi G)$, identically the critical density of a flat FRW universe at every epoch. The cosmological-constant component is the modular-vacuum projection $\rho_\Lambda(z) = \Omega_\Lambda(z) \rho_{\text{info}}(z)$, evaluating at $z = 0$ to $5.84 \times 10^{-27} \text{ kg m}^{-3}$ — the observed scale. The ratio $\rho_\Lambda(0)/\rho_{\text{Pl}} \approx 10^{-123}$ is identified as the inverse of the cosmic diamond's entropy at the present epoch: the 123 orders of magnitude separating the cosmological-constant scale from the Planck scale count the holographic information capacity of the observable universe. The framework predicts a cosmological-constant equation of state $w(z) = -1$ to within 10^{-5} at all observationally relevant redshifts, in tension with the DESI 2024 evolving-dark-energy preference at $2.6\text{--}3.9\sigma$ depending on data combination. Future precision dark-energy surveys provide a direct falsification test: the framework's algebraic structure means the prediction is fixed and cannot be adjusted to accommodate data.

Keywords: cosmological constant; holographic principle; modular structure; causal diamond; dark energy

1 Introduction

The cosmological-constant problem — why the observed dark-energy density $\rho_\Lambda \approx 10^{-123} \rho_{\text{Pl}}$ rather than $\mathcal{O}(1) \rho_{\text{Pl}}$ — has resisted resolution for four decades [1–5]. Standard quantum field theory on a fixed Lorentzian background predicts a vacuum energy density of order ρ_{Pl} from the zero-point modes of the Standard Model fields, in conflict with observation by 123 orders of magnitude. Proposed resolutions include anthropic selection in a string-theory landscape [6, 7], holographic dark-energy bounds with an externally imposed infrared cutoff [8, 9], modified gravity at cosmological scales [10], and various symmetry-based cancellation mechanisms [11, 12]. None has produced a parameter-free derivation of the observed scale; the question of what fixes the magnitude of ρ_Λ has remained open.

This Letter presents a derivation of ρ_Λ from the modular structure of the algebra of observables on the cosmic causal diamond. The result is structural: ρ_Λ at the present epoch is fixed, with no free parameters, by the entropy of the cosmic horizon and the geometry of the diamond. The 10^{-123} hierarchy between ρ_Λ and ρ_{Pl} is identified as the inverse of the cosmic diamond's holographic information capacity at $z = 0$ — a counting argument fixed by algebraic structure rather than a fine-tuning.

The construction uses the Holographic Λ CDM (H- Λ CDM) framework [13–15], in which cosmological structure emerges from the modular flow on $\mathcal{A}(\mathcal{D})$ rather than being imposed on a fixed Lorentzian background. The framework's predictive content has been validated in two companion

analyses: the recombination-horizon sound horizon $r_d = 150.71$ Mpc, in agreement with BAO observations [13]; and a parameter-free prediction $H_{\text{local}} = 69.15 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at the NGC 4258 anchor and $H_{\text{local}} = 70.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at the LMC anchor, validated against Carnegie-Chicago Hubble Program TRGB measurements across four supernova photometric systems [14]. The present work extracts the information stress-energy tensor $T_{\mu\nu}^{\text{info}}$ underlying these predictions, computes its modular-vacuum expectation, and identifies the cosmological-constant scale as the third structural output of the same foundation.

2 The modular vacuum on the cosmic causal diamond

The cosmic causal diamond $\mathcal{D}$ is the spacetime region $\mathcal{V}(p, q) = J^+(p) \cap J^-(q)$, defined as the intersection of the causal future of the Big Bang singularity p with the causal past of the cosmic event horizon q [15], bounded by two null sheets meeting at a 2-surface σ of maximal area — the holographic screen [16–18]. The algebra $\mathcal{A}(\mathcal{D})$ of quantum observables supported on $\mathcal{D}$ carries an intrinsic modular flow whose generator K — the modular Hamiltonian — defines a thermodynamic notion of horizon-region energy and entropy [19–25].

For the cosmic diamond, K takes a form structurally identical to those appearing in the first law of black-hole mechanics [26] and in the Gibbons–Hawking derivation of the de Sitter horizon entropy [27],

$$K = 2\pi \int_{\Sigma} T_{\mu\nu}^{\text{info}} \xi^{\mu} d\Sigma^{\nu}, \quad (1)$$

with Σ a Cauchy surface for $\mathcal{D}$, ξ^{μ} the conformal Killing vector preserving the diamond and degenerating on its boundary, and $T_{\mu\nu}^{\text{info}}$ the stress-energy density of the modular state on $\mathcal{A}(\mathcal{D})$. Under the algebraic-primary stance, K is foundational and $T_{\mu\nu}^{\text{info}}$ is determined by the modular state — inverting the standard QFT direction in which $T_{\mu\nu}$ is fundamental and K is derived.

The von Neumann entropy of the modular state is $\langle K \rangle = S_{\text{max}}$, the Bekenstein–Hawking area law for the diamond [28, 29],

$$S_{\text{max}} = \mathcal{A}/(4G\hbar) = \pi c^5/(G\hbar H^2). \quad (2)$$

This is a *derived* property of the modular state rather than an axiom imposed on the Lorentzian geometry; the holographic principle [16–18] is recovered as a consequence rather than postulated.

Vacuum form. The modular vacuum $|\Omega\rangle$ on $\mathcal{D}$ is invariant under the conformal isometries of the diamond. Any rank-2 tensor expectation in this state must therefore be invariant under those isometries, and the only such tensor is $g_{\mu\nu}$,

$$\langle T_{\mu\nu}^{\text{info}} \rangle_{\Omega} = -\rho_{\Lambda} g_{\mu\nu}, \quad (3)$$

the standard cosmological-constant form with equation of state $p = -\rho$. The form (3) is forced by symmetry alone: no fluid ansatz, no fine-tuning, no input beyond the algebraic structure of $\mathcal{A}(\mathcal{D})$. All that remains is to compute the coefficient ρ_{Λ} .

Total horizon stress-energy density. The modular state is thermal at the modular temperature $T_{\text{mod}} = \hbar H/(2\pi)$, the Gibbons–Hawking temperature of the cosmic horizon [27]. The total horizon energy is the entropy times the temperature,

$$E_{\text{horizon}} = S_{\text{max}} T_{\text{mod}} = \frac{\pi c^5}{G\hbar H^2} \cdot \frac{\hbar H}{2\pi} = \frac{c^5}{2GH}.$$

Dividing by the Hubble volume $V_H = (4\pi/3)(c/H)^3$ yields the total information stress-energy density,

$$\rho_{\text{info}}(z) = \frac{E_{\text{horizon}}}{V_H} = \frac{3c^2 H^2(z)}{8\pi G}, \quad (4)$$

the de Sitter horizon energy density [18, 27] recovered here from the cosmic-diamond modular state. By the Friedmann constraint [30], derivable in this framework from the entanglement first law of [31, 32], applied to a flat FRW background [33, 34], this is identically the critical density at every epoch:

$$\rho_{\text{info}}(z) = \rho_{\text{crit}}(z) = \rho_{\Lambda}(z) + \rho_m(z) + \rho_r(z). \quad (5)$$

Equation (5) is a *structural prediction* of the framework: spatial flatness is not assumed as an empirical input but follows from the modular construction. The total horizon stress-energy density automatically saturates the critical-density bound at every epoch, with the partition into vacuum and excitations determined by the matter and radiation content of the Lorentzian state above the modular vacuum.

3 The cosmological-constant scale

The vacuum (Λ) component is the $\Omega_\Lambda(z)$ projection of $\rho_{\text{info}}(z)$,

$$\rho_\Lambda(z) = \Omega_\Lambda(z) \rho_{\text{info}}(z) = \frac{3c^2 \Omega_\Lambda(z) H^2(z)}{8\pi G}, \quad (6)$$

the coefficient appearing in equation (3).

Numerical evaluation at $z = 0$. Using the Planck 2018 best-fit $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1} = 2.184 \times 10^{-18} \text{ s}^{-1}$ and $\Omega_\Lambda(0) = 0.6847$ [35], with $c = 2.998 \times 10^8 \text{ m s}^{-1}$ and $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$:

$$\begin{aligned} \rho_{\text{info}}(0) &= 7.67 \times 10^{-10} \text{ J m}^{-3} = 8.53 \times 10^{-27} \text{ kg m}^{-3} = \rho_{\text{crit}}(0), \\ \rho_\Lambda(0) &= \Omega_\Lambda(0) \rho_{\text{info}}(0) = 5.84 \times 10^{-27} \text{ kg m}^{-3}, \end{aligned}$$

matching the observed cosmological-constant density to within Planck 2018 measurement precision.

The hierarchy. In Planck units $\rho_{\text{Pl}} = c^5/(\hbar G^2) = 5.15 \times 10^{96} \text{ kg m}^{-3}$,

$$\rho_\Lambda(0)/\rho_{\text{Pl}} = 1.13 \times 10^{-123}. \quad (7)$$

The $\sim 10^{-123}$ suppression of the cosmological-constant scale relative to the Planck scale [1] is identified within the framework as the inverse of the cosmic diamond's holographic information capacity:

$$\rho_\Lambda(0)/\rho_{\text{Pl}} \sim \Omega_\Lambda/N_{\text{P}}(0), \quad (8)$$

with $N_{\text{P}}(0) = \mathcal{A}(0)/\ell_{\text{Pl}}^2 \approx 9 \times 10^{122}$ the number of Planck-area cells on the cosmic horizon at $z = 0$. The 123 orders of magnitude separating the observed cosmological constant from the Planck density count the holographic information capacity of the observable universe.

The hierarchy is not a fine-tuning. It is a counting argument fixed by the algebraic structure of $\mathcal{A}(\mathcal{D})$ at the present epoch, with the Ω_Λ factor encoding what fraction of the horizon stress-energy lives in the modular-vacuum (Λ -form) component versus the matter and radiation excitations. No anthropic selection, no string landscape, no holographic IR cutoff is required: the cosmological constant is the modular-vacuum stress-energy of $T_{\mu\nu}^{\text{info}}$ on the cosmic diamond, with magnitude fixed by the diamond's entropy and the Gibbons–Hawking temperature of its boundary.

4 The dark-energy equation of state

The framework's prediction for the dark-energy equation of state $w(z)$ follows from the modular-vacuum coefficient's evolution under modular flow. The vacuum component of $T_{\mu\nu}^{\text{info}}$ obeys the standard continuity equation in the Lorentzian representation,

$$\dot{\rho}_\Lambda + 3H(\rho_\Lambda + p_\Lambda) = 0,$$

with $p_\Lambda = w(z) \rho_\Lambda$. Inverting,

$$w(z) = -1 + \frac{1+z}{3} \frac{d \ln \rho_\Lambda}{dz}. \quad (9)$$

The evolution of $\rho_\Lambda(z)$ under modular flow connects the diamond at $z = 0$ to the diamond at z through proper-time evolution at the holographic information-processing rate $\gamma(z) = H(z)/\ln N_{\text{P}}(z)$ [13, 14], with integrated modular parameter

$$\Delta s(z) = \int_0^z \frac{1}{(1+z') \ln N_{\text{P}}(z')} dz'.$$

The leading-order modular-vacuum coefficient is invariant under its own flow (the modular vacuum is the fixed point of modular evolution), so corrections to ρ_Λ enter at second order in the modular small parameter $\varepsilon \equiv (\gamma/H) \Delta s(z)$,

$$\rho_\Lambda(z) = \rho_\Lambda(0) \left[1 + \mathcal{O}(\varepsilon^2) \right].$$

At $z = 0.5$, $\Delta s(0.5) \approx \ln(1.5)/281.7 \approx 1.4 \times 10^{-3}$ and $\varepsilon \approx 5 \times 10^{-6}$. Translating to $w(z)$ via equation (9),

$$w(z) = -1 + \mathcal{O}(10^{-5}) \quad (10)$$

at all observationally relevant redshifts. The framework predicts a true cosmological constant at the equation-of-state level, with corrections three orders of magnitude below current survey precision.

Tension with DESI 2024. The DESI BAO collaboration's first-year analysis [36], combined with CMB and Type Ia supernova data, reports a $2.6\text{--}3.9\sigma$ preference for evolving dark energy in the Chevallier–Polarski–Linder (CPL) parametrization $w(z) = w_0 + w_a z/(1+z)$, with $w_0 \approx -0.55$ and $w_a \approx -1.7$ in the most significant data combination. The Λ CDM prediction (10) is in direct tension with this fit: $w_0 = -1$, $w_a = 0$ to within 10^{-5} .

This tension is the framework's principal current empirical exposure. Three observations bear on its interpretation. First, the DESI signal's significance is data-combination-dependent: the BAO-only fit reaches 2.6σ ; the higher significance is achieved only when CMB and Pantheon+ supernova data are added [36, 37], which means a fraction of the apparent significance is tension between DESI BAO and other late-time probes rather than a clean BAO measurement of evolving dark energy. The recent SDSS+DESI joint analysis [38] obtains $w_0 = -0.838 \pm 0.055$ and $w_a = -0.62^{+0.22}_{-0.20}$, lower significance than the original DESI Y1 release. Second, future precision dark-energy surveys (DESI Y5, Euclid, Roman) will substantially constrain the systematic-uncertainty floor that currently dominates the CPL fit's data-combination dependence; the question of whether dark energy is genuinely evolving will be sharpened considerably as next-generation precision data becomes available. Third, the Λ CDM prediction's algebraic structure makes it falsifiable in a structurally clean way: the framework cannot accommodate $w_0 \neq -1$, regardless of how the DESI signal evolves with future data. If precision surveys converge on $w(z) \neq -1$, the framework is falsified at the equation-of-state level. If they converge on $w(z) = -1$, the framework is corroborated.

The asymmetry of this falsification structure — a parameter-free prediction either survives or dies, with no parameter-fit middle ground — distinguishes the framework from phenomenological dark-energy models in which $w(z)$ is fitted from the data and the question is which functional form best fits. The Λ CDM prediction is a pre-registered quantitative claim that future observations will adjudicate.

5 Conclusion

The cosmological-constant energy density follows from the modular structure of the cosmic causal diamond's algebra of observables. The total horizon stress-energy density saturates the critical density at every epoch (equation 5), the modular-vacuum projection at $z = 0$ matches the observed Λ scale (equation 6), and the 10^{-123} Planck-density hierarchy is identified as the inverse holographic information capacity of the observable universe (equation 8). The framework predicts a true cosmological constant equation of state $w(z) = -1$ to within 10^{-5} (equation 10), a sharp claim falsifiable against precision dark-energy surveys.

Three structural results emerge from a single foundation: spatial flatness as a structural prediction rather than an empirical input; the cosmological-constant scale as a counting argument rather than a fine-tuning; and a parameter-free $w(z)$ prediction sharper than current observational bounds by three orders of magnitude. The same modular foundation has been used to resolve the BAO sound-horizon tension at recombination [13] and the local-vs-CMB Hubble tension at distance-ladder anchors [14]; the present derivation extends the framework's predictive reach to the cosmological constant. None of these results requires fitting; all three follow from the algebraic structure of $\mathcal{A}(\mathcal{D})$ and the Lorentzian projection of $T_{\mu\nu}^{\text{info}}$.

The framework's empirical exposure is asymmetric. It cannot be adjusted to accommodate evolving dark energy if DESI 2024's preference is confirmed by precision surveys; it cannot be adjusted to

accommodate spatial curvature if future CMB or large-scale-structure measurements detect $\Omega_K \neq 0$ at high significance; it cannot be adjusted to accommodate Hubble tension residuals if precision TRGB measurements diverge from the predictions of [14]. The same algebraic structure that fixes the cosmological-constant scale fixes all of these predictions, making the framework falsifiable at multiple independent observational fronts simultaneously. The empirical record of the next decade — DESI Y5, Euclid, Roman, JWST-anchored TRGB campaigns, DESI Year 3 BAO — will determine whether the modular foundation’s predictions are corroborated across independent tests or falsified at one or more.

The cosmological-constant problem, in this picture, has a structural answer: the magnitude of Λ is what holographic information accounting on the cosmic causal diamond delivers when computed from the modular state. The 123 orders of magnitude are the entropy of the observable universe.

Acknowledgments

[Placeholder]

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